FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°1_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMESINFORMATIQUES I

PRÉNOM: Lens Sandro

NOM: PÉTIOTE

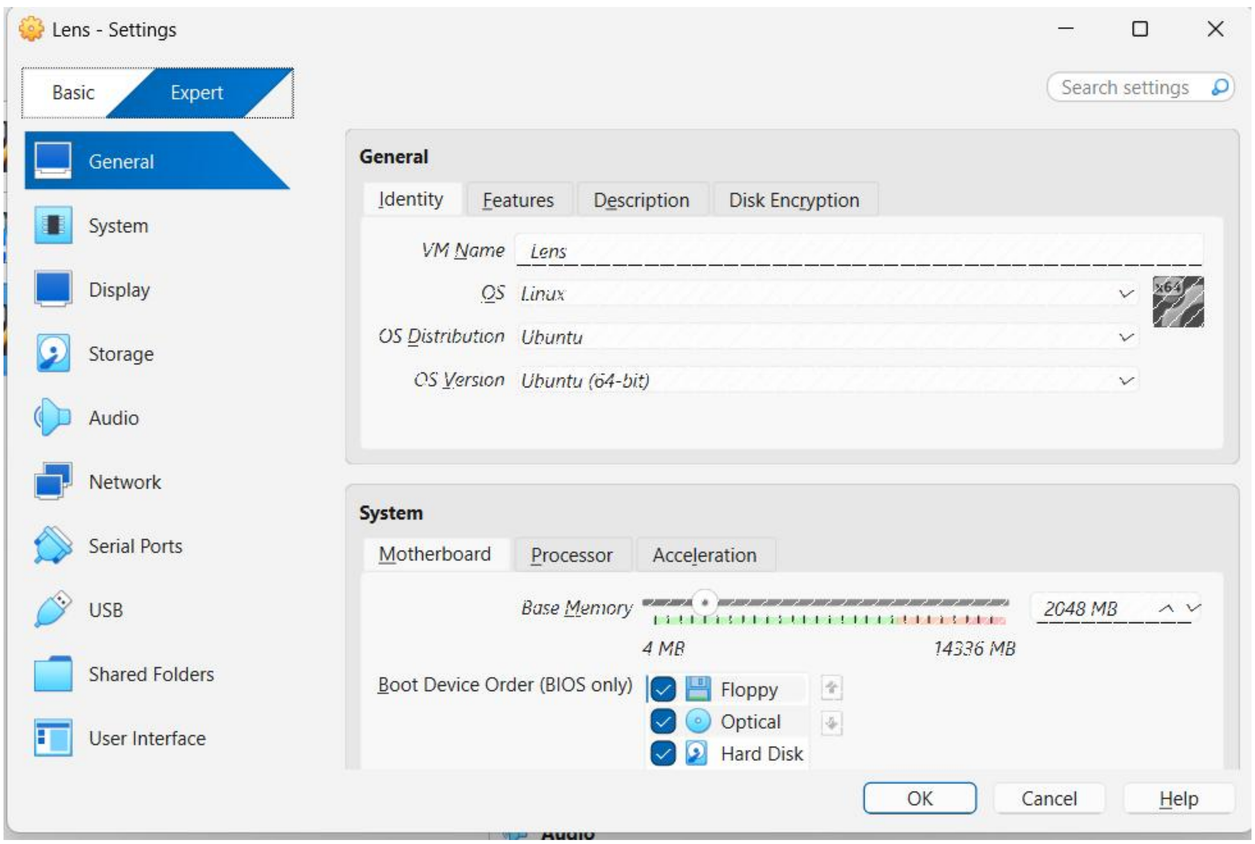
NIVEAU: L3

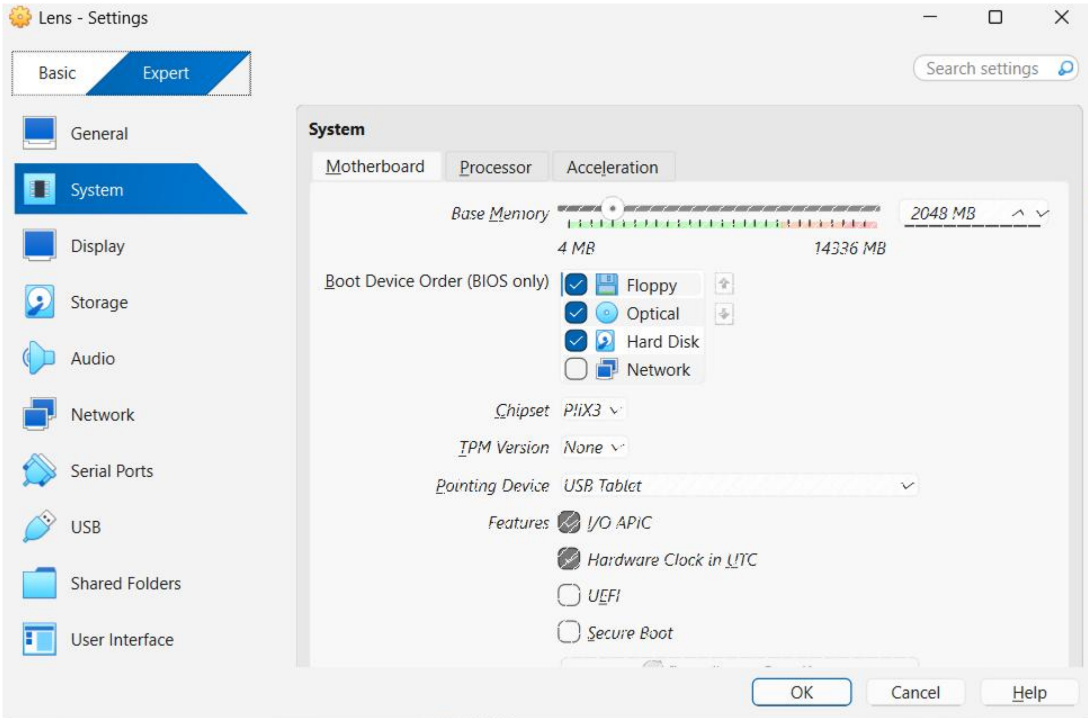
DATE: 07/03/2026

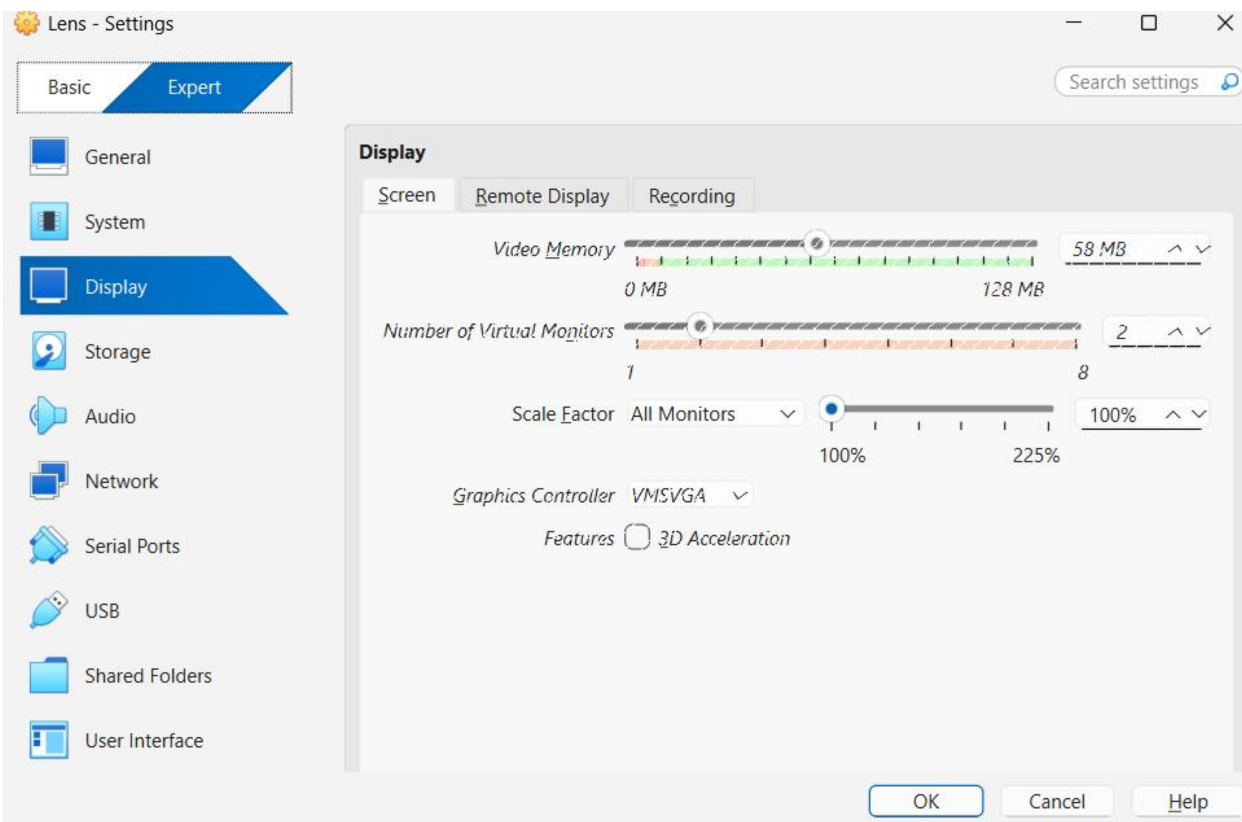
L’objectif de ce TD est de:

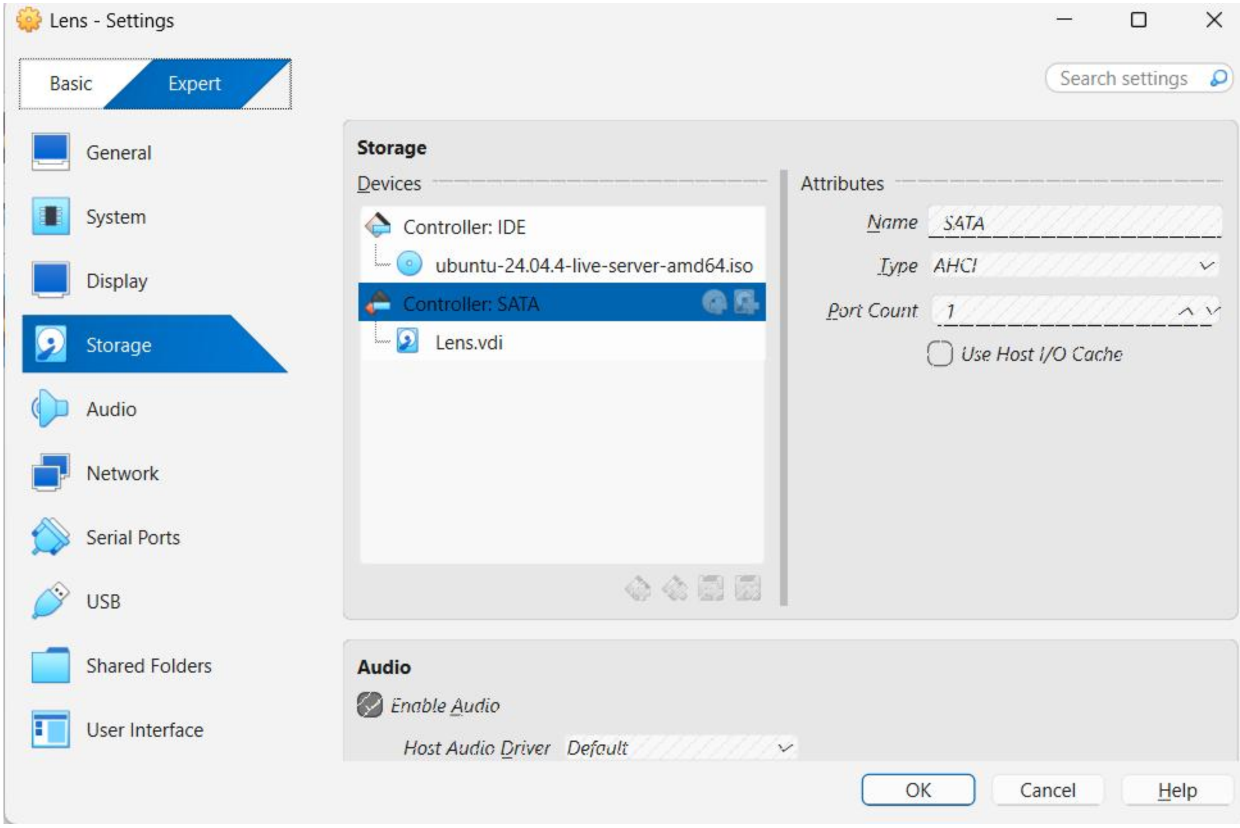
1. Installer un serveur Linux minimal
2. Configurer le réseau
3. Mettre en place l’accès distant sécurisé avec SSH
4. Effectuer la configuration de base d’un serveur

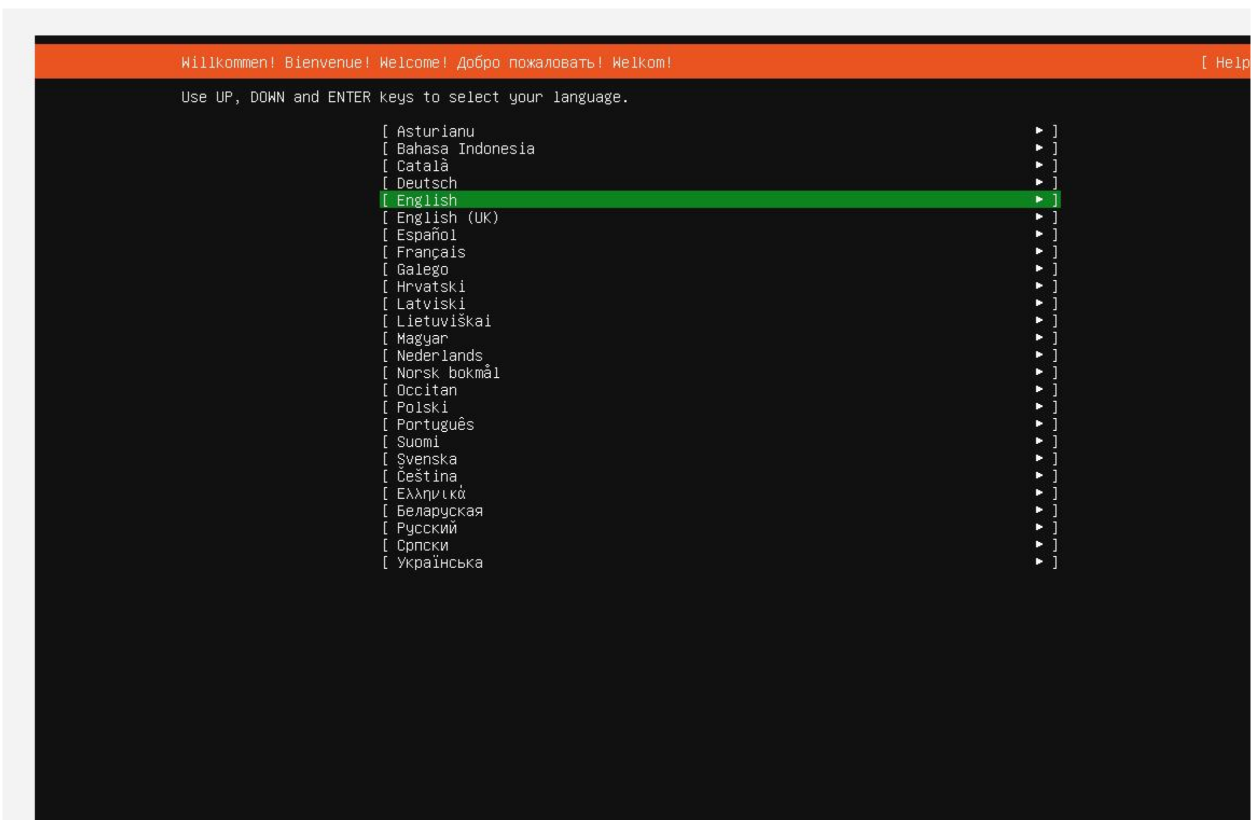
1- Installation de Ubuntu serveur et creation de la machine

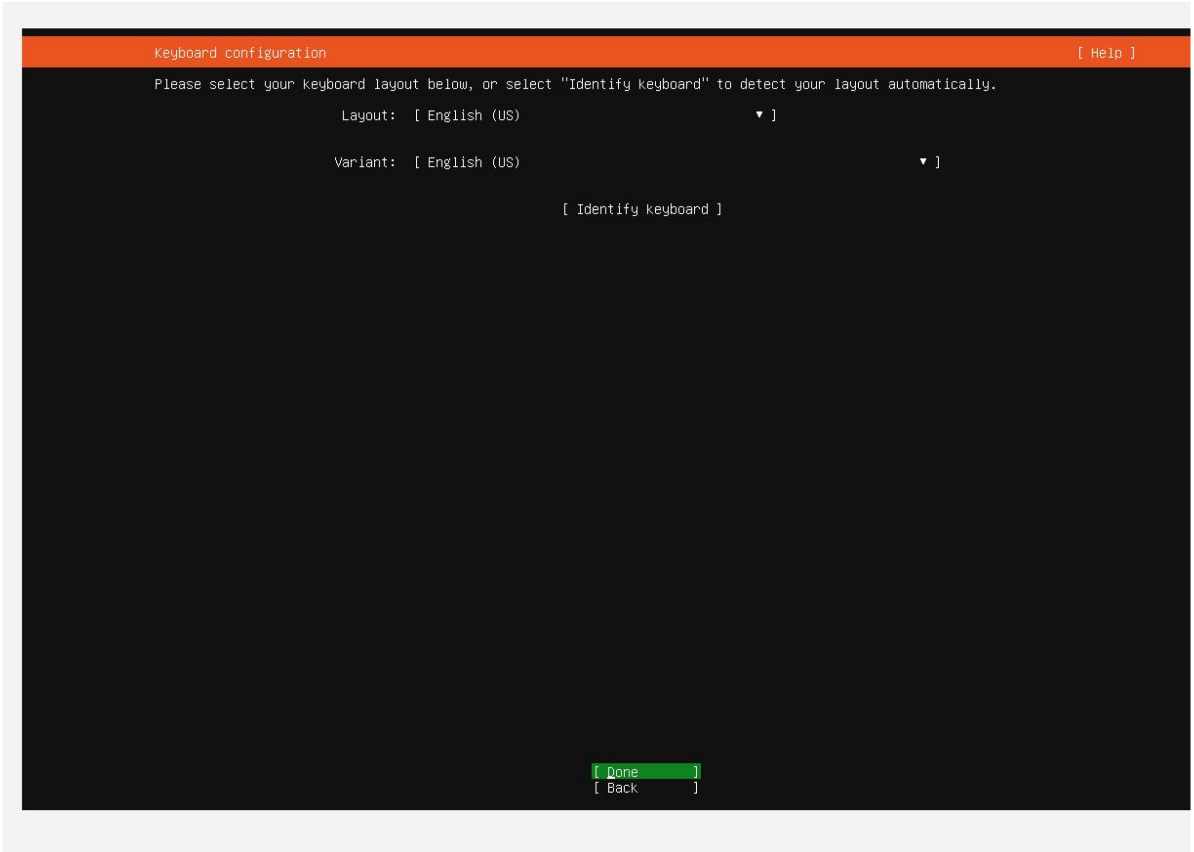


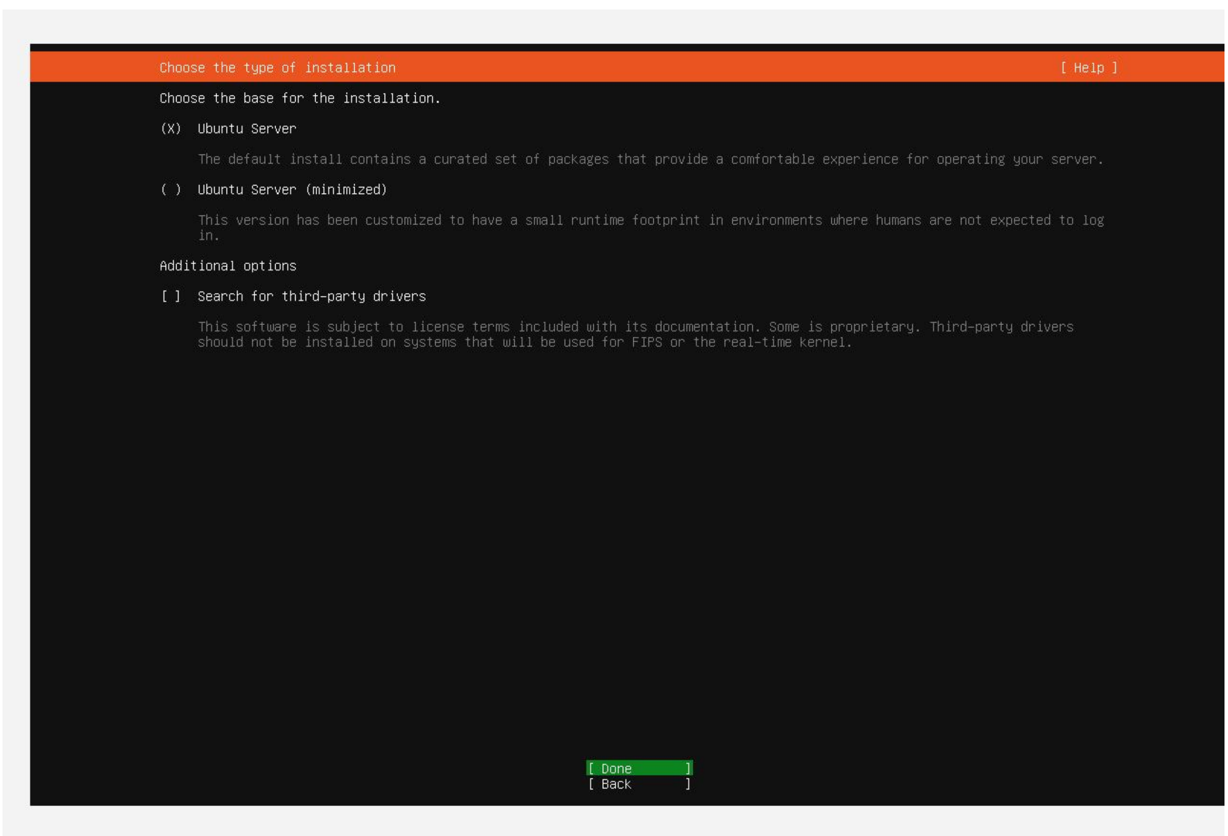

Lens - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
System
Motherboard
Processor
Acceleration
Base Memory
2048 MB
4 MB
14336 MB
Boot Device Order (BIOS only)
Floppy
Optical
Hard Disk
Network
Chipset
PIIX3
TPM Version
None
Pointing Device
USB Tablet
Features
I/O APIC
Hardware Clock in UTC
UEFI
Secure Boot
OK
Cancel
Help

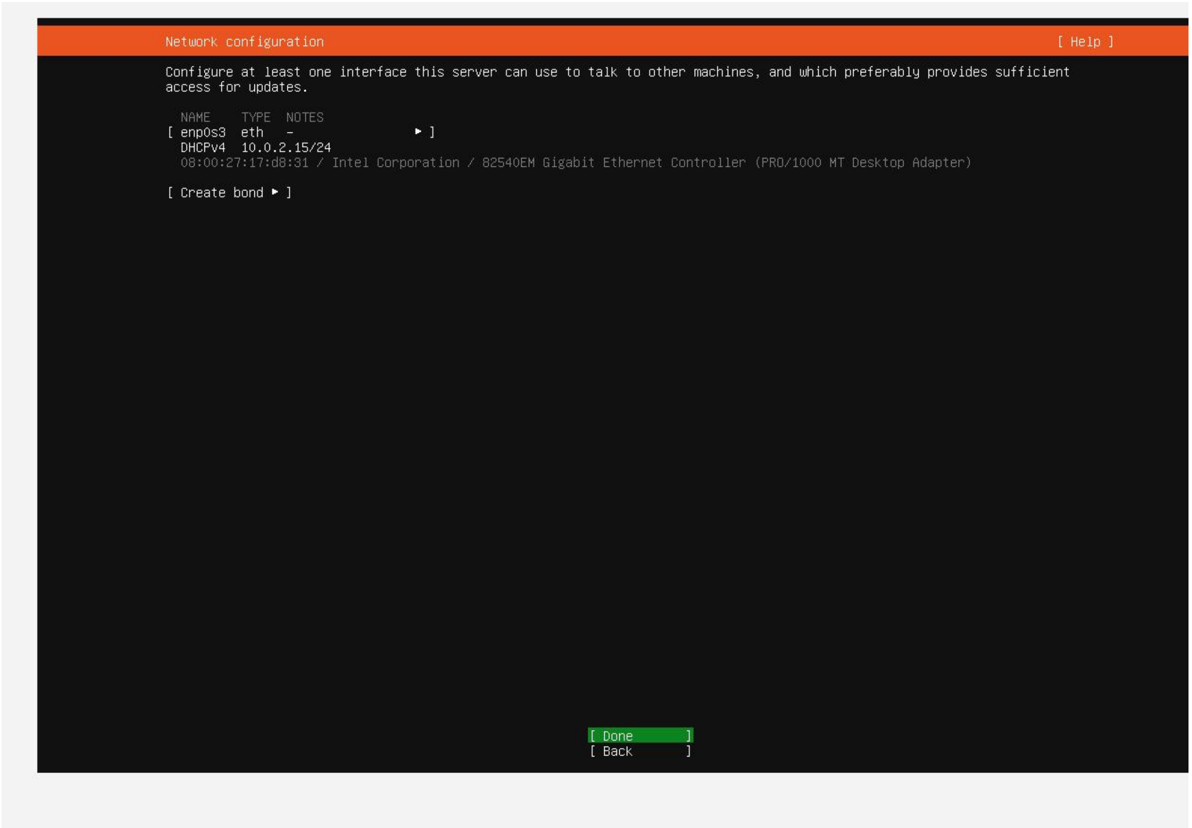

Lens - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Display
Screen
Remote Display
Recording
Video Memory
58 MB
0 MB
128 MB
Number of Virtual Monitors
2
1
8
Scale Factor
All Monitors
100%
100%
225%
Graphics Controller
VMSVGA
Features
3D Acceleration
OK
Cancel
Help

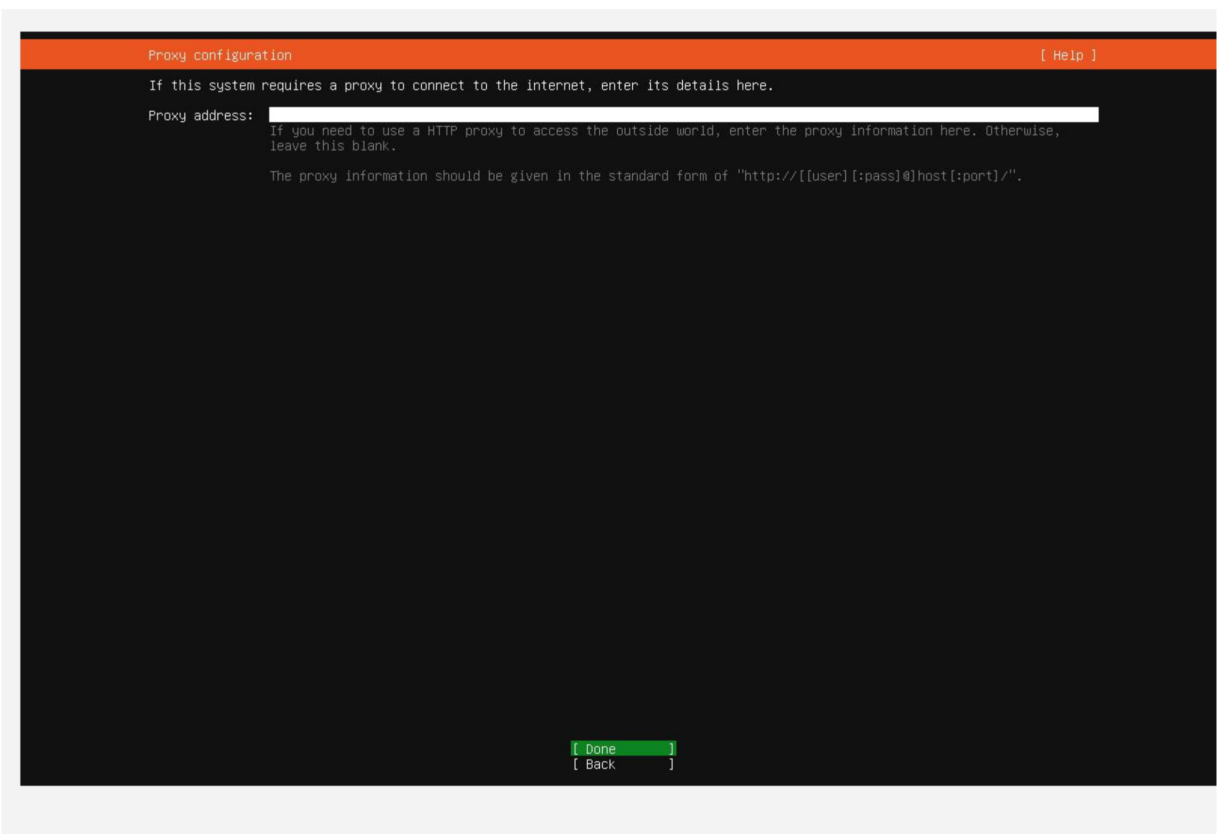

Lens - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Storage
Devices
Controller: IDE
ubuntu-24.04.4-live-server-amd64.iso
Controller: SATA
Lens.vdi
Attributes
Name
SATA
Type
AHCI
Port Count
1
Use Host I/O Cache
Audio
Enable Audio
Host Audio Driver
Default
OK
Cancel
Help

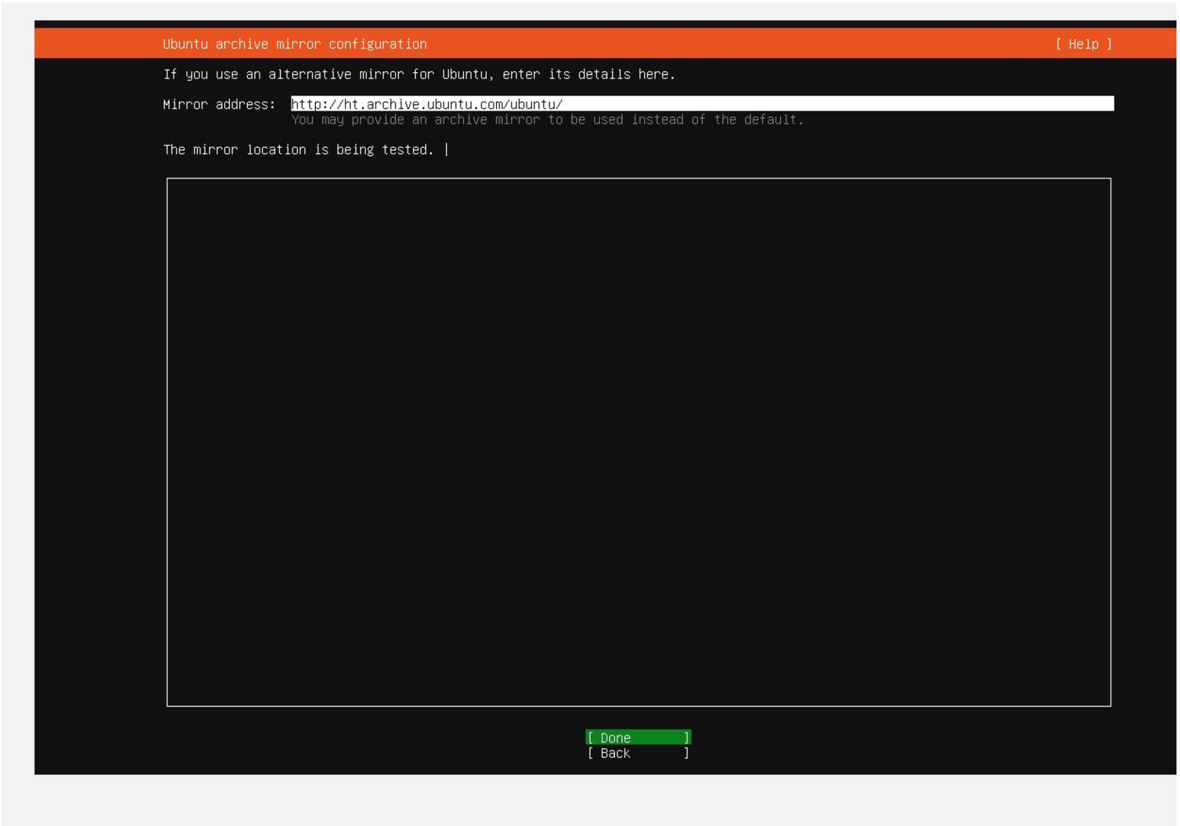

Willkommen! Bienvenue! Welcome! Добро пожаловать! Welkom!
[Help
Use UP, DOWN and ENTER keys to select your language.
[Asturianu ▸]
[Bahasa Indonesia ▸]
[Català ▸]
[Deutsch ▸]
[English ▸]
[English (UK) ▸]
[Español ▸]
[Français ▸]
[Galego ▸]
[Hrvatski ▸]
[Latviski ▸]
[Lietuviškai ▸]
[Magyar ▸]
[Nederlands ▸]
[Norsk bokmål ▸]
[Occitan ▸]
[Polski ▸]
[Português ▸]
[Suomi ▸]
[Svenska ▸]
[Čeština ▸]
[Ελληνικά ▸]
[Беларуская ▸]
[Русский ▸]
[Српски ▸]
[Українська ▸]

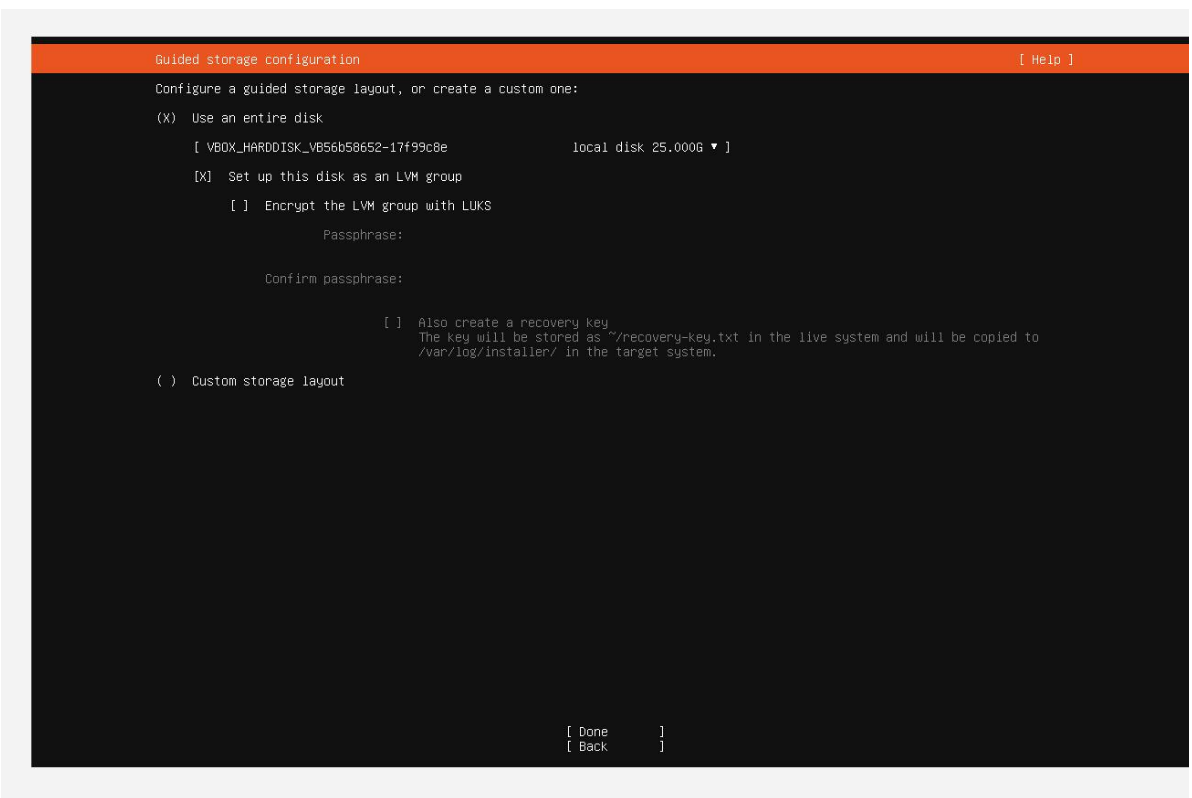

Keyboard configuration
[Help]
Please select your keyboard layout below, or select "Identify keyboard" to detect your layout automatically.
Layout: [English (US) ▼]
Variant: [English (US) ▼]
[Identify keyboard]
[Done]
[Back]

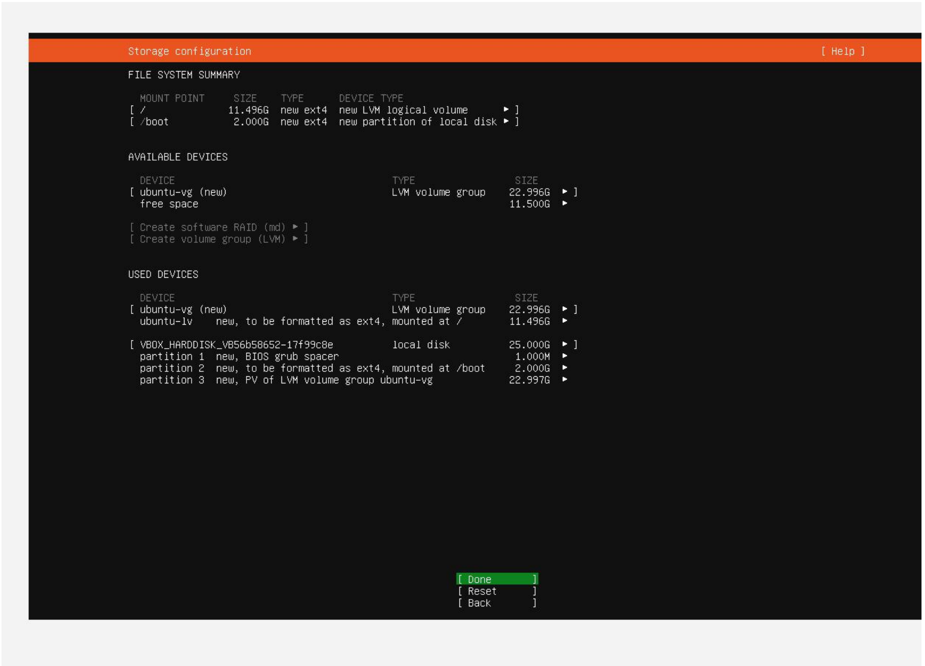

Choose the type of installation
[Help]
Choose the base for the installation.
(X) Ubuntu Server
The default install contains a curated set of packages that provide a comfortable experience for operating your server.
( ) Ubuntu Server (minimized)
This version has been customized to have a small runtime footprint in environments where humans are not expected to log in.
Additional options
[ ] Search for third-party drivers
This software is subject to license terms included with its documentation. Some is proprietary. Third-party drivers should not be installed on systems that will be used for FIPS or the real-time kernel.
[Done]
[Back]


Network configuration
[Help]
Configure at least one interface this server can use to talk to other machines, and which preferably provides sufficient access for updates.
NAME TYPE NOTES
[enp0s3 eth - ►]
DHCPv4 10.0.2.15/24
08:00:27:17:d8:31 / Intel Corporation / 82540EM Gigabit Ethernet Controller (PRO/1000 MT Desktop Adapter)
[Create bond ►]
[Done]
[Back]

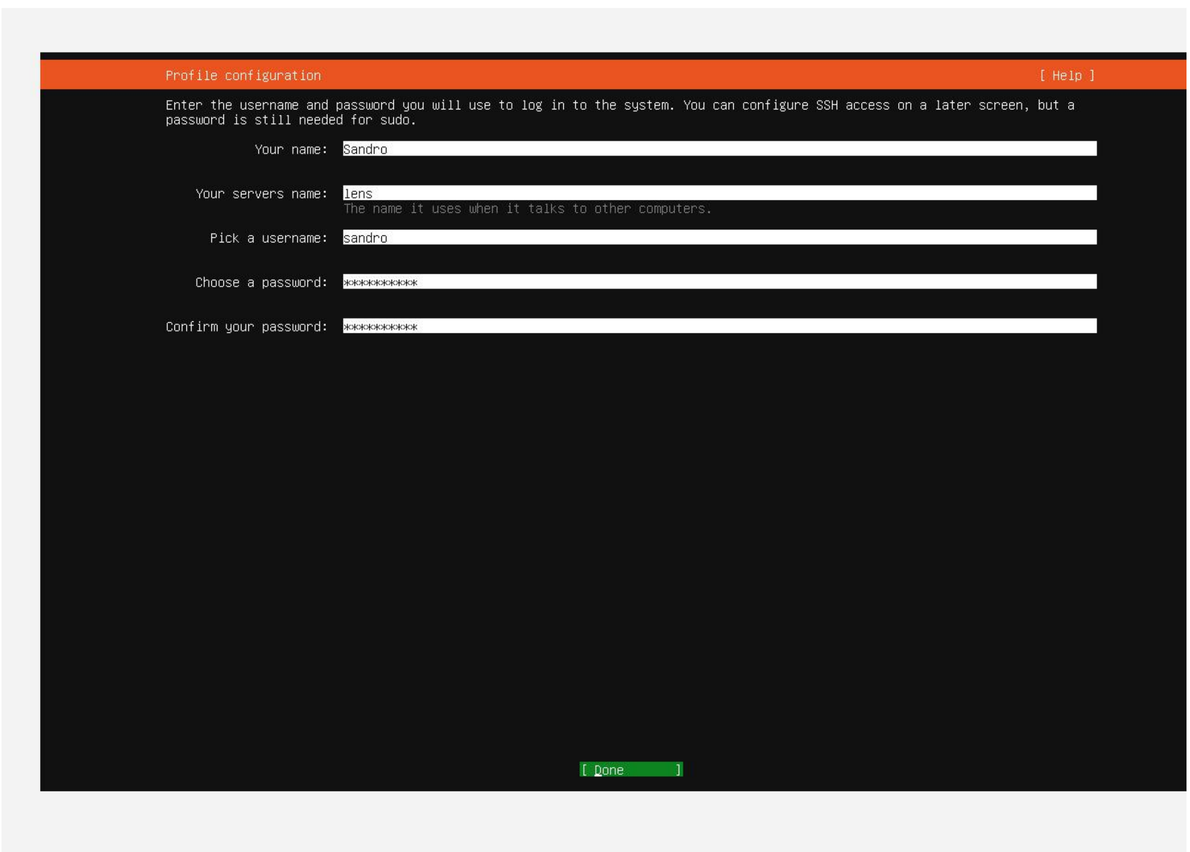

Proxy configuration
[Help]
If this system requires a proxy to connect to the internet, enter its details here.
Proxy address:
If you need to use a HTTP proxy to access the outside world, enter the proxy information here. Otherwise, leave this blank.
The proxy information should be given in the standard form of "http://[[user][:pass]@]host[:port]/".
[Done]
[Back]


Ubuntu archive mirror configuration
[Help]
If you use an alternative mirror for Ubuntu, enter its details here.
Mirror address:
http://ht.archive.ubuntu.com/ubuntu/
You may provide an archive mirror to be used instead of the default.
The mirror location is being tested. |
[Done]
[Back]

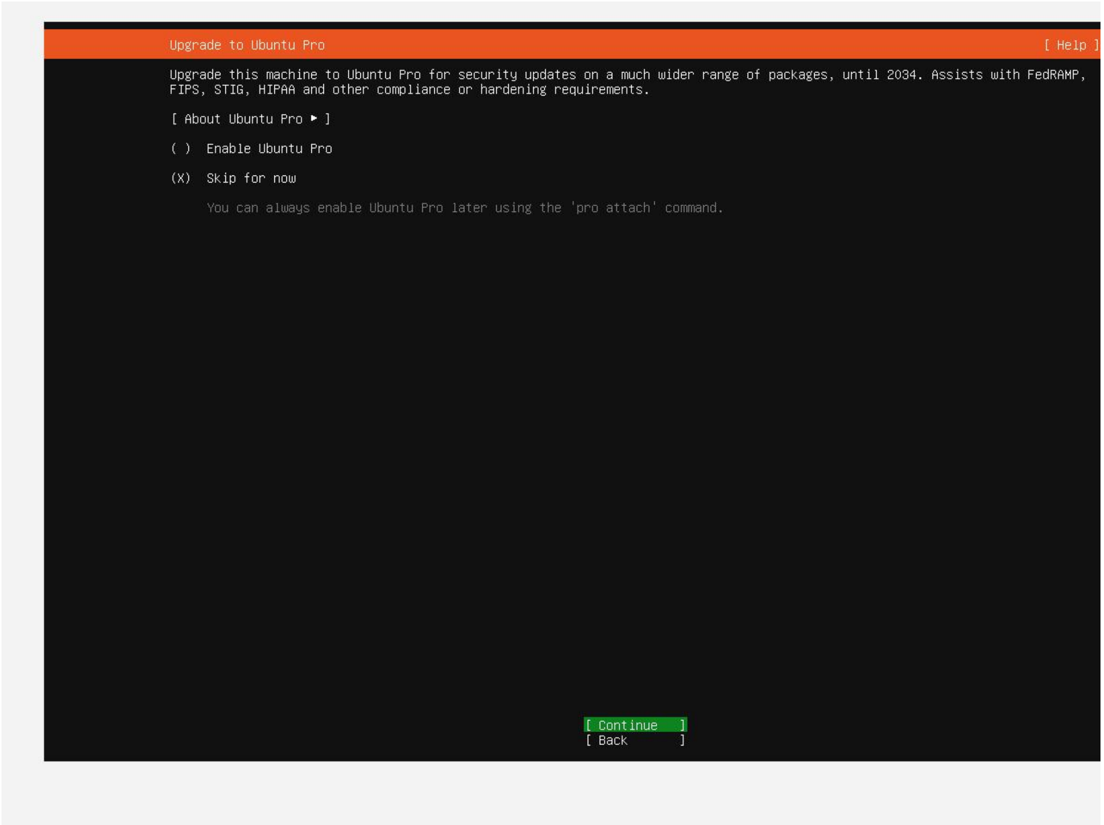

Guided storage configuration
[Help]
Configure a guided storage layout, or create a custom one:
(X) Use an entire disk
[VBOX_HARDDISK_VB56b58652-17f99c8e local disk 25.000G ▾]
[X] Set up this disk as an LVM group
[ ] Encrypt the LVM group with LUKS
Passphrase:
Confirm passphrase:
[ ] Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to
/var/log/installer/ in the target system.
( ) Custom storage layout
[Done]
[Back]

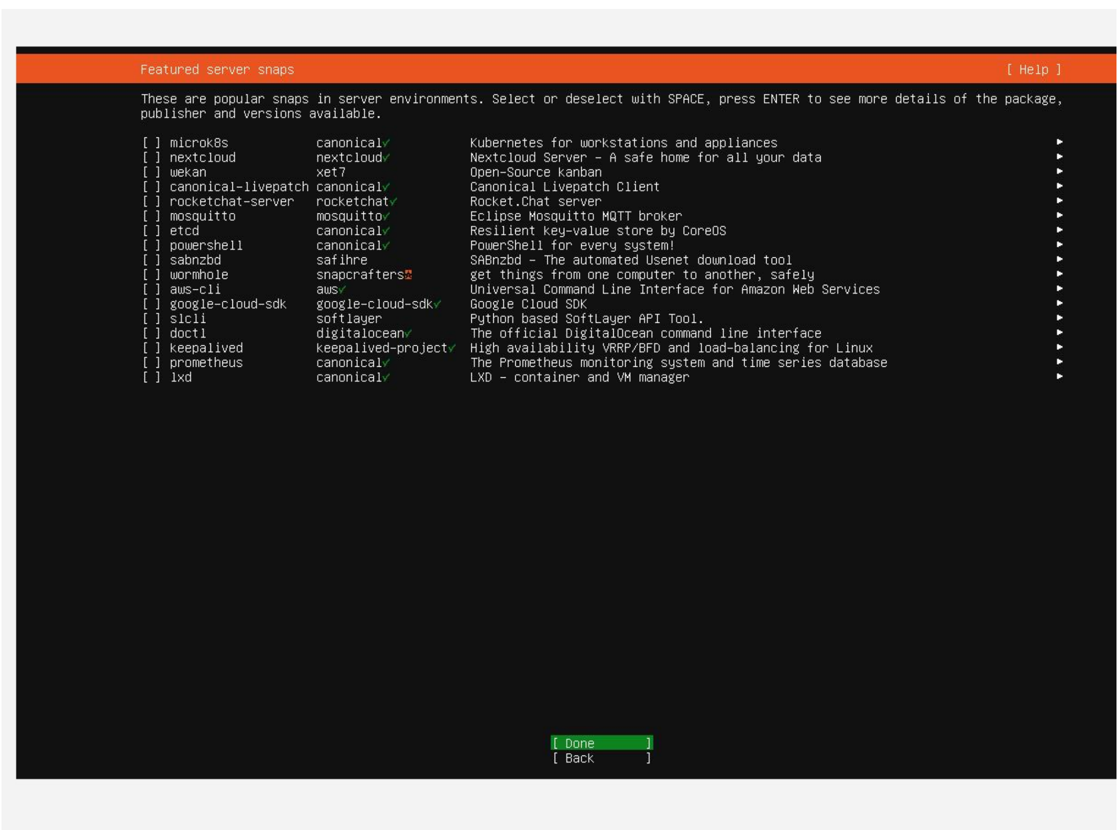


Storage configuration
[Help]
FILE SYSTEM SUMMARY
MOUNT POINT SIZE TYPE DEVICE TYPE
[/ 11.496G new ext4 new LVM logical volume ▸]
[/boot 2.000G new ext4 new partition of local disk ▸]
AVAILABLE DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
free space 11.500G ▸
[Create software RAID (md) ▸]
[Create volume group (LVM) ▸]
USED DEVICES
DEVICE TYPE SIZE
[ubuntu-vg (new) LVM volume group 22.996G ▸]
ubuntu-lv new, to be formatted as ext4, mounted at / 11.496G ▸
[VBOX_HARDDISK_VB56b58652-17f99c8e local disk 25.000G ▸]
partition 1 new, BIOS grub spacer 1.000M ▸
partition 2 new, to be formatted as ext4, mounted at /boot 2.000G ▸
partition 3 new, PV of LVM volume group ubuntu-vg 22.997G ▸
[Done]
[Reset]
[Back]



Profile configuration
[Help]
Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.
Your name: Sandro
Your servers name: lens
The name it uses when it talks to other computers.
Pick a username: sandro
Choose a password: **********
Confirm your password: **********
[Done]


Upgrade to Ubuntu Pro
[Help]
Upgrade this machine to Ubuntu Pro for security updates on a much wider range of packages, until 2034. Assists with FedRAMP, FIPS, STIG, HIPAA and other compliance or hardening requirements.
[About Ubuntu Pro ►]
( ) Enable Ubuntu Pro
(X) Skip for now
You can always enable Ubuntu Pro later using the 'pro attach' command.
[Continue]
[Back]


Featured server snaps
[Help]
These are popular snaps in server environments. Select or deselect with SPACE, press ENTER to see more details of the package, publisher and versions available.
[ ] microk8s canonical✓ Kubernetes for workstations and appliances ►
[ ] nextcloud nextcloud✓ Nextcloud Server - A safe home for all your data ►
[ ] wekan xet7 Open-Source kanban ►
[ ] canonical-livepatch canonical✓ Canonical Livepatch Client ►
[ ] rocketchat-server rocketchat✓ Rocket.Chat server ►
[ ] mosquitto mosquitto✓ Eclipse Mosquitto MQTT broker ►
[ ] etcd canonical✓ Resilient key-value store by CoreOS ►
[ ] powershell canonical✓ PowerShell for every system! ►
[ ] sabnzbd safihre SABnzbd - The automated Usenet download tool ►
[ ] wormhole snapcrafters get things from one computer to another, safely ►
[ ] aws-cli aws✓ Universal Command Line Interface for Amazon Web Services ►
[ ] google-cloud-sdk google-cloud-sdk✓ Google Cloud SDK ►
[ ] slcli softlayer Python based SoftLayer API Tool. ►
[ ] doctl digitalocean✓ The official DigitalOcean command line interface ►
[ ] keepalived keepalived-project✓ High availability VRRP/BFD and load-balancing for Linux ►
[ ] prometheus canonical✓ The Prometheus monitoring system and time series database ►
[ ] lxd canonical✓ LXD - container and VM manager ►
[Done]
[Back]

2- Depuis votre machine physique, exécutez ces commandes via PowerShell.

- Connection à mon serveur depuis ma machine physique et test de quelques commandes.

```
PS C:\Users\Lens Sandro Petiote> ssh lens@172.20.10.4
The authenticity of host '172.20.10.4 (172.20.10.4)' can't be established.
ED25519 key fingerprint is SHA256:nODC/5aoIY4E9BhwtIzMjRr6IoPB7ncus2ADLZNPqBw.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.20.10.4' (ED25519) to the list of known hosts.
lens@172.20.10.4's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Mar  8 06:02:44 PM UTC 2026

  System load:  0.31               Processes:               103
  Usage of /:   40.7% of 11.21GB   Users logged in:         1
  Memory usage: 9%                 IPv4 address for enp0s3: 172.20.10.4
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


lens@lens:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
lens@lens:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:17:d8:31 brd ff:ff:ff:ff:ff:ff
    inet 172.20.10.4/28 metric 100 brd 172.20.10.15 scope global dynamic enp0s3
       valid_lft 86116sec preferred_lft 86116sec
    inet6 fe80::a00:27ff:fe17:d831/64 scope link
       valid_lft forever preferred_lft forever
lens@lens:~$ ping -c 4 google.com
PING google.com (173.194.215.101) 56(84) bytes of data.
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=1 ttl=104 time=92.8 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=2 ttl=104 time=97.1 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=3 ttl=104 time=92.1 ms
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=4 ttl=104 time=87.5 ms
```

```
64 bytes from vt-in-f101.1e100.net (173.194.215.101): icmp_seq=4 ttl=104 time=87.5 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 87.541/92.383/97.067/3.377 ms
lens@lens:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              197M  1.1M  196M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   12G  4.6G  6.1G  43% /
tmpfs                              985M     0  985M   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  104M  1.7G   6% /boot
tmpfs                              197M   12K  197M   1% /run/user/1000
lens@lens:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       318Mi       1.5Gi       1.0Mi       208Mi       1.6Gi
Swap:          2.0Gi          0B       2.0Gi
lens@lens:~$ sudo systemctl status ssh
[sudo] password for lens:
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sun 2026-03-08 17:59:56 UTC; 6min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 700 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 712 (sshd)
      Tasks: 1 (limit: 2263)
     Memory: 4.1M (peak: 5.1M)
        CPU: 341ms
     CGroup: /system.slice/ssh.service
             └─712 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 08 17:59:53 lens systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Mar 08 17:59:56 lens sshd[712]: Server listening on 0.0.0.0 port 22.
Mar 08 17:59:56 lens sshd[712]: Server listening on :: port 22.
Mar 08 17:59:56 lens systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Mar 08 18:02:44 lens sshd[940]: Accepted password for lens from 172.20.10.2 port 64872 ssh2
Mar 08 18:02:44 lens sshd[940]: pam_unix(sshd:session): session opened for user lens(uid=1000) by lens(uid=0)
lens@lens:~$ ps
    PID TTY          TIME CMD
    996 pts/0    00:00:00 bash
   1036 pts/0    00:00:00 ps
lens@lens:~$ top
top - 18:06:57 up 6 min,  2 users,  load average: 0.08, 0.20, 0.14
Tasks:  97 total,   1 running,  96 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  6.2 sy,  0.0 ni, 93.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1968.1 total,   1586.3 free,    317.1 used,    210.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1651.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    995 lens      20   0   15128   7116   5148 S   4.5   0.4   0:02.89 sshd
      1 root      20   0   22116  13248   9476 S   0.0   0.7   0:05.87 systemd
top - 18:06:58 up 6 min,  2 users,  load average: 0.08, 0.20, 0.14
Tasks:  97 total,   2 running,  95 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  2.5 sy,  0.0 ni, 95.0 id,  0.0 wa,  0.0 hi,  2.5 si,  0.0 st
```

- Visualisation de mon systeme en temps réèl avec TOP

```
top - 18:06:57 up 6 min,  2 users,  load average: 0.08, 0.20, 0.14
Tasks:  97 total,   1 running,  96 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  6.2 sy,  0.0 ni, 93.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   1968.1 total,   1586.3 free,    317.1 used,    210.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1651.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    995 lens      20   0   15128   7116   5148 S   4.5   0.4   0:02.89 sshd
      1 root      20   0   22116  13248   9476 S   0.0   0.7   0:05.87 systemd
top - 18:06:58 up 6 min,  2 users,  load average: 0.08, 0.20, 0.14
Tasks:  97 total,   2 running,  95 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  2.5 sy,  0.0 ni, 95.0 id,  0.0 wa,  0.0 hi,  2.5 si,  0.0 st
MiB Mem :   1968.1 total,   1586.3 free,    317.1 used,    210.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1651.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
top - 18:08:07 up 8 min,  2 users,  load average: 0.02, 0.16, 0.13
Tasks:  97 total,   1 running,  96 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.9 sy,  0.0 ni, 98.7 id,  0.0 wa,  0.0 hi,  0.4 si,  0.0 st
MiB Mem :   1968.1 total,   1586.3 free,    317.1 used,    210.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1651.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
     55 root      20   0       0      0      0 I   1.3   0.0   0:04.70 kworker/0:2-events
    995 lens      20   0   15128   7116   5148 S   0.3   0.4   0:03.03 sshd
top - 18:09:17 up 9 min,  2 users,  load average: 0.00, 0.12, 0.11
Tasks:  97 total,   1 running,  96 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  1.3 sy,  0.0 ni, 98.3 id,  0.0 wa,  0.0 hi,  0.4 si,  0.0 st
MiB Mem :   1968.1 total,   1586.3 free,    317.1 used,    210.6 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   1651.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
     55 root      20   0       0      0      0 I   1.0   0.0   0:05.49 kworker/0:2-events
    649 message+  20   0    9780   5484   4660 S   0.3   0.3   0:00.20 dbus-daemon
      1 root      20   0   22116  13248   9476 S   0.0   0.7   0:05.87 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_g
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_p
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
      9 root      20   0       0      0      0 I   0.0   0.0   0:00.56 kworker/0:1-cgroup_destroy
     11 root      20   0       0      0      0 I   0.0   0.0   0:02.36 kworker/u2:0-events_power_effici
     12 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_pe
     13 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.37 ksoftirqd/0
     17 root      20   0       0      0      0 I   0.0   0.0   0:00.20 rcu_preempt
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.01 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kdevtmpfs
     22 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_
     23 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kauditd
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.00 khungtaskd
```

Coclusion.

Ce TD m'a permis de savoir comment installer un serveur Linux minimal, configure le réseau, mettre en place l'accès distant sécurisé avec SSH et effectuer la configuration de base d'un serveur, donc la tache a été réussie**.**